



Iron-catalyzed aerobic oxidation of silyl ethers to carboxylic acids†

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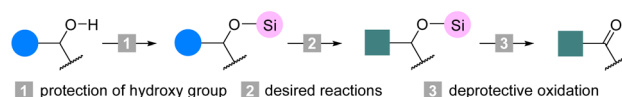
Direct aerobic oxidation of silyl ethers to carboxylic acids has been developed. The mild reaction conditions lead to a broad range of functional group compatibility. Different types of silyl groups have been investigated and selective deprotective oxidation has been realized. The reaction could be conducted under air.

The hydroxy group is a versatile reactive functional group, and silyl groups are commonly used for protecting this functionality aimed at targeted transformations, which would be followed by deprotective oxidation in organic synthesis (Scheme 1A).^{1,2} Catalytic aerobic oxidation has been developed for the deprotective oxidation of silyl ethers to afford aldehydes and ketones.³ The pioneering preparation of carboxylic acids directly from silyl ethers generally lacks efficiency or suffers from functional group tolerance (Scheme 1B).^{4–6} Moreover, apart from TMS and TBS ethers, the performance of other popular silyl ethers in catalyzed aerobic oxidation remains unknown. Herein, we report the iron-catalyzed aerobic oxidation of silyl ethers forming carboxylic acids using TEMPO or 4-OH-TEMPO as a co-catalyst (Scheme 1C). Under mild reaction conditions, a broad range of functional groups such as halogen, ether, tosylate, and ester have been tolerated and the highly selective deprotective oxidation of different silyl ethers has been realized.

By using 1-octyl *tert*-butyldimethylsilylether **1aa** as the model substrate, our investigation began with the evaluation of solvents: with 10 mol% of Fe(NO₃)₃·9H₂O as the catalyst, and TEMPO and KCl as the co-catalysts. Reactions in DCM, DCE, or

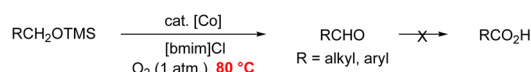
toluene at room temperature for 21 hours hardly or did not give octanoic acid **2a** (Table 1, entries 1–3).⁷ To our delight, the reaction in MeCN or DME gave 82% or 22% NMR yield, respectively (Table 1, entries 4 and 5). The formation of **2a** in 88% NMR yield was detected when using THF as the solvent (Table 1, entry 6). Replacing TEMPO with 4-OH-TEMPO gave a similar yield, while NHPI failed to promote the reaction (Table 1, entries 7 and 8). Considering the cost, 4-OH-TEMPO was chosen as the co-catalyst. Noteworthily, the formation of octyl octanoate in 4% NMR yield was observed in the experiment of entry 7, which must be attributed to the high concentration of the substrate. Running at a lower concentration of **1aa** in THF at 25 °C for 32 hours exclusively gave the desired product **2a** in 92% NMR yield (Table 1, entry 9). No product was detected using FeCl₃ or FeCl₃·6H₂O as the catalyst; among the

A. application of silyl protecting group in organic synthesis

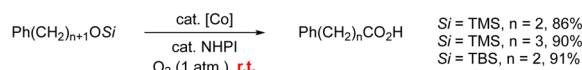


B. previous reports: catalyzed aerobic oxidation of silyl ethers

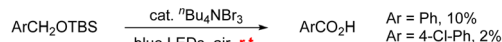
(a) Ahmad Shaabani, 2008



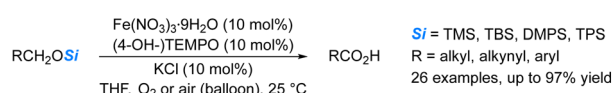
(b) Babak Karimi, 2004



(c) Foad Kazemi, 2020



C. this work: iron-catalyzed aerobic oxidation of silyl ethers to obtain carboxylic acids



Scheme 1 Deprotective oxidation of silyl ethers.

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Table 1 Optimization of the reaction conditions^a

$ \begin{array}{c} \text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O (10 mol\%)} \\ \text{co-catalyst (10 mol\%)} \\ \text{KCl (10 mol\%)} \\ \text{solvent (1.2 mL)} \\ \text{O}_2 \text{ (balloon), time, r.t.} \end{array} \xrightarrow{\quad} $					
Entry	Co-catalyst	Solvent	Time (h)	Yield of 2a ^b (%)	Recovery of 1aa ^b (%)
1	TEMPO	DCM	21	4	56
2	TEMPO	DCE	21	0	100
3	TEMPO	Toluene	21	0	100
4	TEMPO	MeCN	21	82	0
5	TEMPO	DME	21	22	48
6	TEMPO	THF	21	88	0
7	4-OH-TEMPO	THF	21	88 (84 ^c)	0
8	NHPI	THF	21	0	57
9 ^d	4-OH-TEMPO	THF	32	92	0
10 ^e	4-OH-TEMPO	THF	21	0	100
11 ^f	—	THF	21	0	41
12 ^g	4-OH-TEMPO	THF	21	60	8

^a Reaction conditions: **1aa** (5.0 mmol), Fe(NO₃)₃·9H₂O (10 mol%), co-catalyst (10 mol%), KCl (10 mol%), and O₂ (balloon) in solvent (1.2 mL) at room temperature. ^b Determined by ¹H NMR analysis using CH₂Br₂ as the internal standard. ^c Isolated yield. ^d Reaction was conducted using **1aa** (1.0 mmol) in 1.0 mL of THF at 25 °C. ^e Without Fe(NO₃)₃·9H₂O. ^f Without 4-OH-TEMPO. ^g Without KCl.

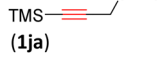
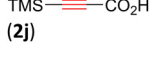
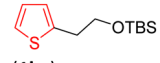
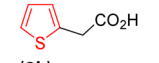
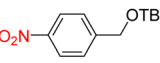
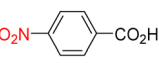
nitrate, only Fe(NO₃)₃·9H₂O and Bi(NO₃)₃·5H₂O worked and Fe(NO₃)₃·9H₂O was more efficient (for details, see ESI† Table S1). Control experiments showed the necessity of both Fe(NO₃)₃·9H₂O and 4-OH-TEMPO (Table 1, entries 10 and 11). In the absence of KCl, a lower yield of **2a** was obtained (Table 1, entry 12).

To uncover the nature of this deprotective oxidation, different silyl ethers were first evaluated. TBS and TMS ethers, which were frequently used in organic synthesis, smoothly went through deprotective oxidation and gave octanoic acid **2a** in 84% yield and 77% yield, respectively (Table 2, entries 1 and 2). The reactions of DMPS or TPS ethers are much slower, giving octanoic acid **2a** in 90% or 74% yield, respectively (Table 2, entries 3 and 4). Substrates with a halogen (**1da**, **1ea**), ether (**1fa**, **1ga**), tosylate (**1ha**), ester (**1ia**), carbon-carbon triple bond (**1ja**), and thienyl (**1ka**) were all tolerated, affording the corresponding carboxylic acids **2d–2k** in moderate to good yields (Table 2, entries 7–14). In the presence of TEMPO, *tert*-butyldimethyl-(4-nitrobenzyl-oxy)silane **1la** was transformed to 4-nitrobenzoic acid **2l** successfully in a mixed solvent (Table 2, entry 15).

TBDPS and TIPS ethers were rather stable and mostly recovered after 59 hours under the standard reaction conditions (Table 3, entries 1 and 2). In substrates with more than one type of silyl groups, TMS, TBS, DMPS, and TPS ethers underwent the deprotective oxidation, while TBDPS or TIPS groups in the same molecule remained untouched (Table 3, entries 3–8). This deprotective selectivity suggested the potential value of this protocol in organic synthesis.

Using TEMPO as the co-catalyst, the deprotective oxidation of silyl ethers could also be carried out in air. Substrates containing 3-phenylpropyl (**1ca**), ether (**1ga**), a carbon-carbon triple bond (**1ja**), and thienyl (**1ka**) were tolerated and gave yields up to 97% (Table 4, entries 1–4). The selective deprotective

Table 2 The substrate scope^a

$ \begin{array}{c} \text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O (10 mol\%)} \\ \text{4-OH-TEMPO (10 mol\%)} \\ \text{KCl (10 mol\%)} \\ \text{O}_2 \text{ (balloon), THF, time, 25 }^\circ\text{C} \end{array} \xrightarrow{\quad} $				
Entry	Substrate	Product	Time (h)	Yield (%)
1 ^b	<i>n</i> -C ₈ H ₁₇ OTBS (1aa)	<i>n</i> -C ₇ H ₁₅ CO ₂ H (2a)	21	84
2	<i>n</i> -C ₈ H ₁₇ OTMS (1ab)	(2a)	24	77
3	<i>n</i> -C ₈ H ₁₇ ODMPS (1ac)	(2a)	50	90
4	<i>n</i> -C ₈ H ₁₇ OTPS (1ad)	(2a)	59	74
5 ^{cd}	<i>n</i> -C ₁₅ H ₃₁ OTBS (1ba)	<i>n</i> -C ₁₄ H ₂₉ CO ₂ H (2b)	45	89
6	Ph(CH ₂) ₃ OTBS (1ca)	Ph(CH ₂) ₂ CO ₂ H (2c)	69	70
7	Cl(CH ₂) ₆ OTBS (1da)	Cl(CH ₂) ₅ CO ₂ H (2d)	64	74
8	Br(CH ₂) ₉ OTBS (1ea)	Br(CH ₂) ₈ CO ₂ H (2e)	71	86
9 ^d	BnO(CH ₂) ₃ OTBS (1fa)	BnO(CH ₂) ₂ CO ₂ H (2f)	53	63
10	<i>n</i> -C ₆ H ₁₃ O(CH ₂) ₂ OTBS (1ga)	<i>n</i> -C ₆ H ₁₃ OCH ₂ CO ₂ H (2g)	63	69
11	TsO(CH ₂) ₂ OTBS (1ha)	TsOCH ₂ CO ₂ H (2h)	60	57
12 ^d	MeO ₂ C(CH ₂) ₅ OTBS (1ia)	MeO ₂ C(CH ₂) ₄ CO ₂ H (2i)	35	75
13	 (1ja)	 (2j)	65	59
14	 (1ka)	 (2k)	70	37
15 ^e	 (1la)	 (2l)	60	76

^a Reaction conditions: **1** (1.0 mmol), Fe(NO₃)₃·9H₂O (10 mol%), 4-OH-TEMPO (10 mol%), KCl (10 mol%), and O₂ (balloon) in THF (1.0 mL) at 25 °C. ^b Reaction was conducted using **1aa** (5.0 mmol) in THF (1.2 mL) at room temperature. ^c Fe(NO₃)₃·9H₂O (20 mol%) was used. ^d 4-OH-TEMPO (20 mol%) was used. ^e TEMPO (10 mol%) was used, with THF (0.5 mL) and DCE (0.5 mL) as the solvent.

oxidation of **3aeb** could also be realized, forming **4ae** in 73% yield (Table 4, entry 5).

Table 3 Selective deprotective oxidation^a

$ \begin{array}{c} \text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O (10 mol\%)} \\ \text{4-OH-TEMPO (10 mol\%)} \\ \text{KCl (10 mol\%)} \\ \text{O}_2 \text{ (balloon), THF, time, 25 }^\circ\text{C} \end{array} \xrightarrow{\quad} $				
Entry	Substrate	Product	Time (h)	Yield (%)
1	<i>n</i> -C ₈ H ₁₇ OTBDPS (1ae)	—	59	0 (99 ^b)
2	<i>n</i> -C ₈ H ₁₇ OTIPS (1af)	—	59	0 (92 ^b)
3	TBDPSO(CH ₂) ₆ OTBS (3aea)	TBDPSO(CH ₂) ₅ CO ₂ H (4ae)	42	74
4	TBDPSO(CH ₂) ₆ OTMS (3aeb)	(4ae)	69	87
5	TBDPSO(CH ₂) ₆ ODMPS (3aec)	(4ae)	47	67
6	TBDPSO(CH ₂) ₆ OTPS (3aed)	(4ae)	91	65
7	TIPSO(CH ₂) ₆ ODMPS (3afc)	TIPSO(CH ₂) ₅ CO ₂ H (4af)	49	48
8	TIPSO(CH ₂) ₅ ODMPS (3bfc)	TIPSO(CH ₂) ₄ CO ₂ H (4bf)	64	58

^a Reaction conditions: **1** or **3** (1.0 mmol), Fe(NO₃)₃·9H₂O (10 mol%), 4-OH-TEMPO (10 mol%), KCl (10 mol%), and O₂ (balloon) in THF (1.0 mL) at 25 °C. ^b Recovery of **1** was determined by ¹H NMR analysis with CH₂Br₂ as an internal standard.

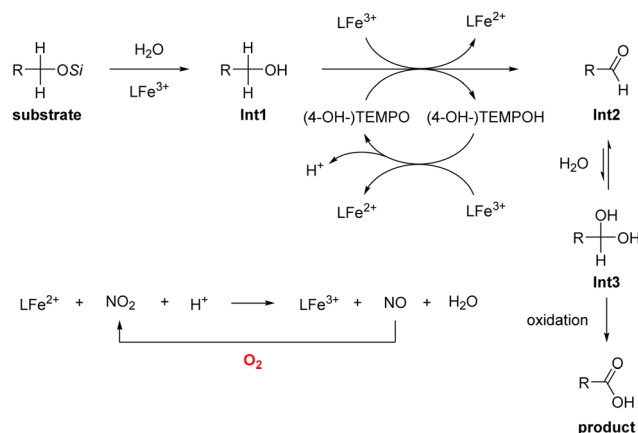
Table 4 Deprotective oxidation in air^a

RCH ₂ OSi (1) or Si ^t O(CH ₂) _n OSi ^t (3)		Fe(NO ₃) ₃ ·9H ₂ O (10 mol%) TEMPO (10 mol%) KCl (10 mol%) air (balloon), THF, time, 25 °C	RCO ₂ H (2) or Si ^t O(CH ₂) _{n-1} CO ₂ H (4)		
Entry	Substrate	Product	Time (h)	Yield (%)	
1	Ph(CH ₂) ₃ OTBS (1ca)	Ph(CH ₂) ₃ CO ₂ H (2c)	94	97 (63 ^b)	
2	<i>n</i> -C ₆ H ₁₃ O(CH ₂) ₂ OTBS (1ga)	<i>n</i> -C ₆ H ₁₃ OCH ₂ CO ₂ H (2g)	72	70 (72 ^b)	
3			47	80	
4			110	58	
5	TBDPSO(CH ₂) ₆ OTMS (3aeb)	TBDPSO(CH ₂) ₅ CO ₂ H (4ac)	24	73	

^a Reaction conditions: **1** or **3** (1.0 mmol), Fe(NO₃)₃·9H₂O (10 mol%), TEMPO (10 mol%), KCl (10 mol%), and air (balloon) in THF (1.0 mL) at 25 °C. ^b 4-OH-TEMPO instead of TEMPO was used and the NMR yield was determined by ¹H NMR analysis with CH₂Br₂ as an internal standard.

Formal syntheses of bioactive molecules **10**⁸ and **11**⁹ were achieved to demonstrate the practicality of this deprotective oxidation protocol. Cyclization of phenol **5** with methyl 3-methyl-2-butenolate (**6**) in methanesulfonic acid gave lactone **7** in 52% yield. The reaction of **7** with lithium aluminum hydride quantitatively afforded the phenol-alkanol **8**. TBS was chosen to protect the alcoholic hydroxy group of **8**. The esterification of the phenolic hydroxyl group of **9** with acetic anhydride or the corresponding acyl chloride formed **1ma** and **1na**, which were conveniently transformed to the carboxylic acids **2m** and **2n** by this deprotective oxidation in 70% and 60% yields, respectively. When TEMPO was used instead of 4-OH-TEMPO, the desired products **2m** and **2n** were both obtained in higher yields (Scheme 2).

Based on our previous study,⁷ we tentatively propose a plausible mechanism (Scheme 3): the silyl ether is hydrolyzed in the presence of Fe³⁺ and H₂O to produce the alcohol **Int1**, which was observed in

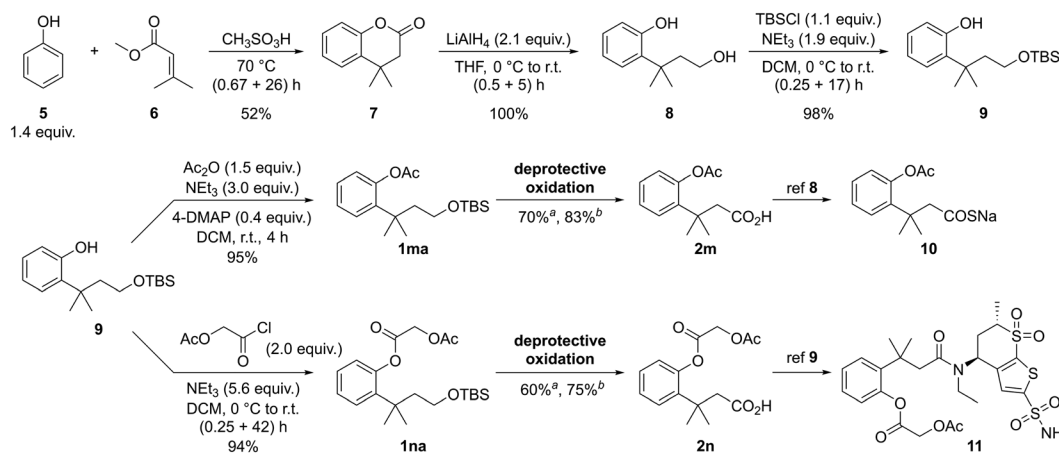


Scheme 3 Plausible mechanism.

the control experiment (Table 1, entry 11). **Int1** would be oxidized by Fe³⁺ and (4-OH)-TEMPO to obtain the aldehyde **Int2** with the formation of Fe²⁺ and (4-OH)-TEMPOH. The hydrate of **Int2**, **Int3**, is further oxidized to the carboxylic acid. (4-OH)-TEMPOH could convert back to (4-OH)-TEMPO in its reaction with Fe³⁺. The Fe²⁺ would be oxidized back to Fe³⁺ by NO₂ affording NO, which would further react with O₂ to regenerate NO₂.

In summary, a Fe(NO₃)₃·9H₂O and (4-OH)-TEMPO-catalyzed aerobic deprotective oxidation has been developed. Highly selective deprotective oxidation of different silyl ethers has also been uncovered: TMS, TBS, DMPS, and TPS ethers underwent the deprotective oxidation; while TBDPS or TIPS ethers in the same molecule did not. The synthetic potential of this protocol has been demonstrated through the formal syntheses of bioactive molecules **10** and **11**. Further studies on the mechanism and applications are ongoing in our laboratory.

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Scheme 2 Formal syntheses of bioactive molecules **10** and **11**. ^a Reaction conditions: **1** (1.0 mmol), Fe(NO₃)₃·9H₂O (10 mol%), 4-OH-TEMPO (10 mol%), KCl (10 mol%), THF (1.0 mL) and O₂ (balloon) at 25 °C, NMR yield. ^b TEMPO (10 mol%) was used instead of 4-OH-TEMPO, isolated yield.

Conflicts of interest

There are no conflicts to declare.

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